Guidelines & Rules 📜

The challenge happens 100% online across two epic stages. Read up on the mechanics, how you will be scored, and the ground rules to make sure your team is ready to dominate! 🚀

🗺️ The Game Plan (Competition Structure)

🛠️ Stage 1: The Build (Asynchronous)

Deadline: 25 March 2026

Submit your project via the official portal:

- 📄 **Tech Report (Max 5 pages):** Detail your problem statement, dataset justification, ML/DL methodology, preprocessing steps, results, interpretability, and clinical impact.
- 🎥 **Video Pitch (Max 3 mins):** Sell your solution! Explain how your mathematical/computational solution brings "hope" to healthcare.
- 💻 **Code Repo (Recommended):** Link your GitHub/GitLab with reproducible code so we can review your magic.

**Judges will shortlist the Top 5 teams based on technical rigor, innovation, interpretability, and clinical relevance.*

🔥 Stage 2: The Grand Final (Synchronous)

Date: 11 April 2026

The Top 5 teams will battle it out live!

- 🌐 **Format:** Live Pitch via Zoom/Webex.
- ⏱️ **Duration:** 10 mins presentation + 5 mins Q&A.
- 🎯 **Focus:** Prove your model's clinical applicability, safety, interpretability, reproducibility, and real-world impact.

📊 How to Win (Evaluation Criteria)

Wondering how the judges will score your project? Here is the exact breakdown of the 100-point rubric:

Criterion	Weight (%)
🧮 **Mathematical/Algorithmic Rigor**	25
📈 **Model Performance & Validation**	20
🧠 **Problem Definition & Healthcare Relevance**	15
💡 **Innovation & Creativity**	15
🛡️ **Clinical Interpretability & Safety**	10
🌟 **"Hope" Impact & Practical Implications**	10
🗣️ **Presentation Quality & Clarity**	5
Total Score	**100**

⚖️ The Ground Rules

Play fair and code hard. Ensure your team follows these regulations to avoid disqualification:

1. 🚫 **No Plagiarism:** Your project must be 100% original work created specifically for this challenge.
2. 📂 **Strict Dataset Compliance:** You *must* exclusively use one (or more) of the official datasets from the PhysioNet list provided (bigP3BCI, Brugada-HUCA, Myocardial Perfusion SPECT, or HYGD). External labeled datasets are strictly not permitted.
3. 🤖 **Ethical AI Only:** Your models need to adhere to strict ethical principles:
   - Prioritize model interpretability and explainability (e.g., saliency maps, feature importance).
   - Include safety validations and consider potential harms.
   - Include an honest discussion of your model's limitations, data leakage prevention, and failure modes.
   - Plan for responsible deployment.
4. 🔓 **Open Source (Encouraged):** While optional, we highly encourage sharing your code for transparency and reproducibility.
5. ⏰ **No Late Submissions:** The Stage 1 deadline is absolute. Submissions received after the cutoff will not be reviewed.
6. 👩‍⚖️ **Judges' Word is Final:** The decision of the judging panel cannot be contested.
7. 🤝 **Squad Loyalty:** Your team composition must remain exactly the same from registration through the finals.

> 📚 **Mandatory Citation Rule:** All submissions must properly cite the specific dataset(s) used (using the official text on each dataset's page) **AND** include the standard PhysioNet

> platform citation. Failure to cite open-source data properly may result in point deductions!

💻 Tech Specs & Tools

Bring your favorite stack! You have complete freedom over your tech, but here is what we recommend based on the datasets:

- **Programming Languages:** Python 🐍 (highly recommended), R, MATLAB, or Julia.
- **Core Scientific Stack:** NumPy, Pandas, SciPy, scikit-learn, PyTorch, TensorFlow, or Keras.
- **Physiological Signal Tooling (EEG/ECG):** * `WFDB` (for ECG in WFDB format)
  - `MNE-Python` (for EEG/ERP workflows)
- **Medical Imaging Tooling (DICOM/NIfTI):**
  - `pydicom` (for DICOM images)
  - `nibabel` / `SimpleITK` (for NIfTI and medical image IO)
  - `MONAI` (Deep learning framework for medical imaging - optional but powerful)
- **Hardware & Environments:** You are in charge of your own compute! Free cloud platforms like Google Colab or Kaggle Notebooks are perfect for initial prototyping. Deep learning on medical images (SPECT/Fundus) may benefit from GPU acceleration. *Pro-tip: Use `requirements.txt` or `conda` environments to ensure your submitted code is easily reproducible!*

📚 Official Learning Resources

Get familiar with the data before you start coding:

- **The Platform:** PhysioNet
- **Dataset 1 (EEG):** bigP3BCI Documentation
- **Dataset 2 (ECG):** Brugada-HUCA Documentation
- **Dataset 3 (SPECT):** Myocardial Perfusion SPECT Documentation
- **Dataset 4 (Fundus):** HYGD (Glaucoma) Documentation

Tip: Brush up on segmentation validation methodologies (Dice score, Hausdorff distance) if tackling imaging, and signal robustness best practices if analyzing EEG/ECG!